

TASK 16: #51629

1. Learn deeply about the different type of logs in linux systems.also learn the real time monitoring of logs as well as the journals.

Note: Logs are found under /var/log/

2.install and configure commonly using firewalls such as CSF and APF. try to manage different services via the same.

3.learn the management of incoming and and outgoing services to an aws instance with security groups.

4.learn about external server attacks and their prevention parameters.

eg: phishing attack, bruteforce attack etc.

1. Logs (/var/log/)

Log File	What it is
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syslog / messages	General system activity — everything happening on the OS.
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auth.log / secure	Login attempts, sudo use, SSH — most important for security.
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kern.log	Kernel messages (drivers, hardware).
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cron	Records of scheduled jobs running.
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maillog	Mail server activity.
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apache2/ or httpd/	Website visits and errors.
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nginx/	Same as above, for Nginx.
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mysql/	Database errors and slow queries.
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btmp / wtmp	Failed and successful login history.
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audit/audit.log	Detailed security audit trail.
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Real-time monitoring:

tail -f /var/log/auth.log # watch a log live

multitail file1 file2 # watch multiple logs live, split screen

lnav /var/log/*.log # smart log viewer, auto-colors and filters

Journal (systemd logs):

```
journalctl -f          # follow live, all logs
journalctl -u sshd -f   # follow just one service
journalctl --since "1 hour ago" # logs from last hour
journalctl -p err       # only errors
```

2. Firewalls — CSF & APF

Both just manage the built-in Linux firewall (iptables) for you, with easier commands.

CSF (recommended, more modern):

```
csf -a <ip> "note"  # allow an IP
csf -d <ip> "note"  # block an IP
csf -r              # apply changes
csf -l              # show current rules
```

To open a new port for a service → edit /etc/csf/csf.conf, add the port to TCP_IN, then run csf -r.

TO Set up them :

Setup CSF

```
sudo dnf update -y
```

Updates all installed packages to their latest versions.

```
sudo dnf install -y perl perl-libwww-perl.noarch wget curl
```

Installs Perl and download tools that CSF needs to run.

```
sudo systemctl stop firewalld
```

Stops RHEL's default firewall so it doesn't conflict with CSF.

```
sudo systemctl disable firewall
```

Prevents firewalld from starting again automatically on reboot.

```
cd /usr/src
```

Moves into the standard directory used for downloading and building software.

```
sudo rm -rf csf* AfazTech-csf*
```

Deletes any leftover CSF files from a previous attempt, to avoid conflicts.

```
sudo wget https://github.com/AfazTech/csf/archive/refs/heads/main.tar.gz -O csf.tar.gz
```

Downloads CSF's installation files from the maintained GitHub fork.

```
sudo tar -xzf csf.tar.gz
```

Extracts the downloaded compressed file.

```
cd csf-main
```

Enters the folder that was just extracted.

```
sudo sh install.sh
```

Runs CSF's installer script, which sets everything up.

```
sudo csf -v
```

Confirms CSF installed correctly by showing its version number.

```
sudo perl /usr/local/csf/bin/csftest.pl
```

Checks that your server supports everything CSF needs to function properly.

```
sudo csf -a $(curl -s ifconfig.me) "my IP"
```

Adds your own current IP to the allow list, so you don't lock yourself out.

```
sudo nano /etc/csf/csf.conf
```

Opens CSF's main configuration file for editing.

Inside the file, set:

```
TESTING = "0"
```

```
TCP_IN = "22,80,443"
```

Turns off the temporary safety mode and defines which ports are allowed in.

```
sudo csf -r
```

Applies your configuration changes to the live firewall.

```
sudo csf -l
```

Displays the current active firewall rules, to confirm everything applied.

Setup APF (only after disabling CSF)

```
sudo systemctl stop csf
```

Stops CSF from running.

```
sudo systemctl stop lfd
```

Stops CSF's brute-force detection service.

```
sudo systemctl disable csf
```

Prevents CSF from starting again automatically on reboot.

```
sudo systemctl disable lfd
```

Prevents LFD from starting again automatically on reboot.

```
sudo csf -x
```

Clears out all of CSF's firewall rules so they don't interfere with APF.

```
sudo dnf install -y git
```

Installs Git, needed to download APF's source code.

```
cd /usr/src
```

Moves into the standard software directory again.

```
sudo git clone https://github.com/rfxn/advanced-policy-firewall.git
```

Downloads APF's installation files from its official repository.

```
cd advanced-policy-firewall
```

Enters the newly downloaded folder.

```
sudo ./install.sh
```

Runs APF's installer script.

```
sudo apf -v
```

Confirms APF installed correctly by showing its version number.

```
sudo nano /etc/apf/conf.apf
```

Opens APF's main configuration file for editing.

Inside the file, set:

```
IG_TCP_CPORTS="21,22,25,53,80,443"
```

```
DEVEL_MODE="0"
```

Defines which ports are allowed in and turns off the temporary safety mode.

```
sudo apf -r
```

Applies your configuration changes to the live firewall.

```
sudo apf -l
```

Displays the current firewall status, to confirm everything applied

Built-in brute-force protection (LFD) auto-blocks IPs after repeated failed logins — configured in the same file.

APF: older, similar idea, fewer features. Same commands with apf instead of csf (apf -a, apf -d, apf -r).

⚠ Never run both together — pick one.

3. AWS Security Groups

A virtual firewall around each AWS instance.

- **Inbound rules** = what traffic is allowed *in* (e.g., port 22 for SSH, 443 for HTTPS).
- **Outbound rules** = what's allowed *out* (default: everything).
- **Stateful** = if you allow a request in, the reply out is automatic — no extra rule needed.

Basic rules to set:

- SSH (22) → only your own IP, never "anywhere."
- HTTP/HTTPS (80/443) → open to everyone.
- Database port (3306, etc.) → only allow traffic from your app server's security group, not the public.

allow SSH only from your IP

```
aws ec2 authorize-security-group-ingress --group-id sg-xxxx --protocol tcp --port 22 --cidr <your-ip>/32
```

Golden rule: only open what you actually need.

4. Common Attacks & How to Prevent Them

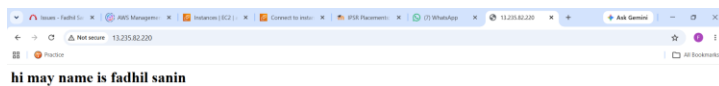
Attack	What it is	How to prevent
Brute-force	Repeated login guessing (SSH, FTP, admin panels).	Use SSH keys instead of passwords, enable CSF/LFD or fail2ban to auto-block after failed attempts.

Attack	What it is	How to prevent
Phishing	Fake emails/sites tricking users into giving up passwords.	Set up SPF/DKIM/DMARC on email, use MFA, train users to spot fakes.
DDoS	Flooding the server with traffic to crash it.	Use AWS Shield/CloudFront, limit connections per IP with CSF or Nginx.
SQL Injection	Malicious input tricking a database into leaking/changing data.	Use parameterized queries in code, add a Web Application Firewall (WAF).
Port Scanning	Attacker probing which ports are open.	Close unused ports, let CSF auto-block scanners.
Privilege Escalation	Attacker gains admin/root after breaking in.	Disable root SSH login, run regular rootkit scans (rkhunter), keep system patched.

Simple rule of thumb: layer your defenses — AWS Security Group → CSF firewall → fail2ban → application-level checks. No single layer should be your only protection.

OUTPUTS:

CFS



APF

